5.

1. In Fig. 1, z is normal to AB. Two parallel rays enter plane AB at C, C', with incidence angle i. The refraction angle is r. The two refracted rays reach the surface at D, D', at incidence angles i_2, i'_2. In $\triangle OCD$, the distance from O to entry C is y_c. By the sine rule,

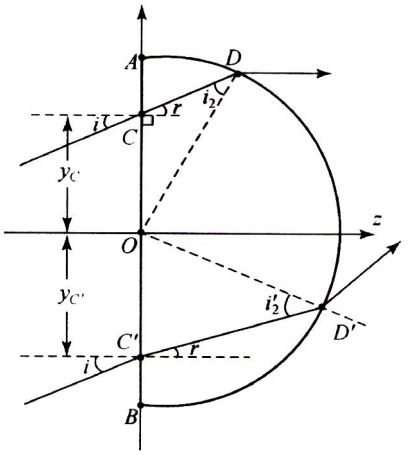


Fig. 1

$$\frac{y_c}{\sin i_2} = \frac{R}{\sin(90^\circ + r)}\ ,\ \text{so}\quad y_c = \frac{\sin i_2}{\cos r}R\ . \qquad (1)$$

Likewise, in $\triangle OC'D'$, the distance from O to entry C' satisfies

$$\frac{y_{c'}}{\sin i'_2} = \frac{R}{\sin(90^\circ - r)}\ ,\ \text{so}\quad y_{c'} = \frac{\sin i'_2}{\cos r}R\ . \qquad (2)$$

Varying i changes refraction r and surface incidence i_2, i'_2. At the critical angle given by

$$i_{20} = \arcsin(1/n) \approx 41.8^\circ \qquad (3)$$

total reflection occurs. Substitute $i_2 = i'_2 = i_{20}$ into (1), (2) to obtain

$$y_{oc} = y_{oc'} = \frac{\sin i_{20}}{\cos r}R\ , \qquad (4)$$

Thus

$$d = 2y_{oc} = 2\frac{\sin i_{20}}{\cos r}R\ . \qquad (5)$$

If $y_c > y_{oc}$ and $y_{c'} > y_{oc'}$, the rays enter, but reach the curved surface at angles above the critical angle i_{20}, so cannot exit. r increases with i. From (4), at $r \to 0$, i.e. $i \to 0$ (normal incidence), d is minimal:

$$d_{\min} = 2R\sin i_{20} = 1.33\,R\ . \qquad (6)$$

At $i \to 90^\circ$ (grazing), $r \to 41.8^\circ$. Substitute $r = 41.8^\circ$ into (4) to find $d_{\max} = 1.79\ \ R$. (7)

2. In Fig. 2, φ is the Oz - OD angle; φ' is the Oz - OD' angle. At total reflection,

$$\varphi = i_{20} + r\ , \qquad (8)$$

$$\varphi' = i_{20} - r\ , \qquad (9)$$

and

$$\theta = \varphi + \varphi' = 2i_{20} \approx 83.6^\circ\ . \qquad (10)$$

Thus θ is independent of i, i.e. θ of i. At grazing

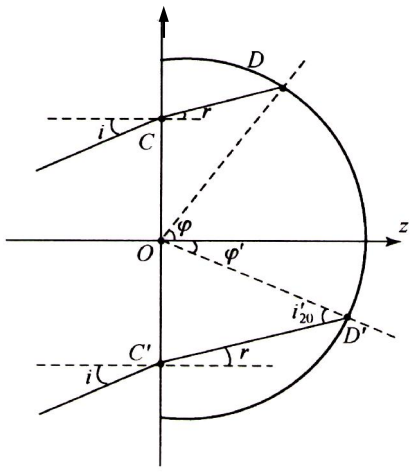


Fig. 2